

Mathematics for the Slightly Rusty but Still Ambitious

A Practical Review for Interviews, Work, and Future Moments of Panic

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Chapter 1

Multivariable Calculus

1.1 Limits, Continuity, and Derivatives

Definition 1.1.1: Function

A function $f : D \rightarrow E$ is a rule that assigns to each element $x \in D$ exactly one element $f(x) \in E$. The set D is the *domain*, and the set of all actual output values is the *range*.

Increasing and Decreasing Functions Let f be defined on an interval I :

- Whenever $x_1 < x_2$ in I , f is *increasing* on I if $f(x_1) < f(x_2)$, is *nondecreasing* on I if $f(x_1) \leq f(x_2)$.
- Whenever $x_1 < x_2$ in I , f is *decreasing* on I if $f(x_1) > f(x_2)$, is *nonincreasing* on I if $f(x_1) \geq f(x_2)$.

Informal Definition of Limit The statement $\lim_{x \rightarrow a} f(x) = L$ means that $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a , from both the left and the right, but with $x \neq a$.

Right- and Left-Hand Limits $\lim_{x \rightarrow a^+} f(x) = L$ means that $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a with $x > a$. Similarly, $\lim_{x \rightarrow a^-} f(x) = L$ means that x approaches a from the left, so $x < a$.

Two-Sided Limit The two-sided limit exists and equals L if and only if:

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L$$

Vertical Asymptote The line $x = a$ is a *vertical asymptote* of the graph of f if at least one of the following is true:

$$\lim_{x \rightarrow a^-} f(x) = \pm\infty, \quad \lim_{x \rightarrow a^+} f(x) = \pm\infty$$

Theorem 1.1.2: Squeeze Theorem

Suppose $g(x) \leq f(x) \leq h(x)$ for all x near a . If $\lim_{x \rightarrow a} g(x) = L$ and $\lim_{x \rightarrow a} h(x) = L$, then $\lim_{x \rightarrow a} f(x) = L$.

Definition 1.1.3: Formal Definition of Limit

Let f be defined near a , except possibly at a . We say $\lim_{x \rightarrow a} f(x) = L$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $0 < |x - a| < \delta$, we have $|f(x) - L| < \varepsilon$.

Continuity at a Point A function $f : D \rightarrow \mathbb{R}$ is *continuous at* $a \in D$ if $\lim_{x \rightarrow a} f(x) = f(a)$.

Continuity on an Interval A function is *continuous on an interval* if it is continuous at every point of that interval. At endpoints, continuity is understood using the appropriate one-sided limit.

Types of Discontinuity. Typical discontinuities include:

- *removable discontinuity*: the limit exists but $f(a)$ is missing or has the wrong value;
- *jump discontinuity*: the one-sided limits exist but are not equal;
- *infinite discontinuity*: the function tends to $\pm\infty$;
- *oscillatory discontinuity*: the function oscillates and has no limiting value.

Theorem 1.1.4: Intermediate Value Theorem

Suppose f is continuous on the closed interval $[a, b]$. If N is any number between $f(a)$ and $f(b)$, then there exists at least one $c \in [a, b]$ such that $f(c) = N$.

Limit at Infinity $\lim_{x \rightarrow +\infty} f(x) = L$ means that $\forall \varepsilon > 0$, there exists $N > 0$ such that if $x > N$, then $|f(x) - L| < \varepsilon$. Similarly, $\lim_{x \rightarrow -\infty} f(x) = L$ means that $\forall \varepsilon > 0$, there exists $N > 0$ such that if $x < -N$, then $|f(x) - L| < \varepsilon$.

Horizontal Asymptote The line $y = L$ is a *horizontal asymptote* if

$$\lim_{x \rightarrow +\infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

Definition 1.1.5: Tangent Line

If the limit $m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists, then the tangent line to the graph of f at $(a, f(a))$ is

$$y - f(a) = m(x - a)$$

Definition 1.1.6: Derivative at a Point

The derivative of f at a is

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

The derivative is the slope of the tangent line and the instantaneous rate of change of f at a .

Derivative Function The derivative of f is the function defined at all x where this limit exists:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

Theorem 1.1.7: Differentiability Implies Continuity

If f is differentiable at a , then f is continuous at a .

The converse is false. A function can be continuous but not differentiable. $f(x) = |x|$.

Basic Differentiation Rules Let $c \in \mathbb{R}$ and suppose f, g are differentiable. Then

$$\frac{d}{dx}[c] = 0 \quad \frac{d}{dx}[x^n] = nx^{n-1} \quad \frac{d}{dx}[cf(x)] = cf'(x) \quad \frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

Theorem 1.1.8: Product Rule and Quotient Rule

If f and g are differentiable, then fg is differentiable and

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x) \quad \frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Derivative Formulas for Trigonometric Functions

$$\begin{aligned} \frac{d}{dx} \sin x &= \cos x & \frac{d}{dx} \cos x &= -\sin x & \frac{d}{dx} \tan x &= \sec^2 x \\ \frac{d}{dx} \cot x &= -\csc^2 x & \frac{d}{dx} \sec x &= \sec x \tan x & \frac{d}{dx} \csc x &= -\csc x \cot x \end{aligned}$$

Theorem 1.1.9: Chain Rule

If g is differentiable at x and f is differentiable at $g(x)$, let $y = f(u)$ and $u = g(x)$, then

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x) \quad \frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Theorem 1.1.10: Inverse Function Derivative

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

Exponential and Logarithm Derivatives

$$\begin{aligned} \frac{d}{dx} e^x &= e^x & \frac{d}{dx} a^x &= a^x \ln a \\ \frac{d}{dx} \ln x &= \frac{1}{x} & \frac{d}{dx} \log_a x &= \frac{1}{x \ln a} \end{aligned}$$

Inverse Trigonometric Derivatives

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}} \quad \frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}} \quad \frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

Implicit Derivative A function may be defined implicitly by a relation between x and y ; instead of solving for y explicitly, differentiate both sides with respect to x , treating y as a function of x , and apply the chain rule when differentiating expressions involving y , e.g. $\frac{d}{dx}[y] = y'$, $\frac{d}{dx}[y^2] = 2yy'$, $\frac{d}{dx}[\sin y] = \cos(y)y'$.

1.2 Applications of Derivatives

Linearization If f is differentiable at $x = a$, the **linearization** of f at a is $L(x) = f(a) + f'(a)(x - a)$.

Differential If $y = f(x)$, then $dy = f'(x) dx$. The actual change is $\Delta y = f(x + \Delta x) - f(x) \approx dy = f'(x)\Delta x, \Delta x \rightarrow 0$.

Theorem 1.2.1: Extreme Value Theorem

If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum and minimum on $[a, b]$.

Local and Absolute Extremum

- f has a **local maximum** at c if $f(c) \geq f(x)$ for all x sufficiently close to c .
- f has a **local minimum** at c if $f(c) \leq f(x)$ for all x sufficiently close to c .
- f has an **absolute maximum** at c on a domain D if $f(c) \geq f(x)$ for all $x \in D$.
- f has an **absolute minimum** at c on a domain D if $f(c) \leq f(x)$ for all $x \in D$.

Theorem 1.2.2: Fermat's Theorem

If f has a local maximum or local minimum at c , and if $f'(c)$ exists, then

$$f'(c) = 0$$

This is a necessary condition, not a sufficient condition. $f'(c) = 0$ only makes c a candidate. $f(x) = x^3$.

Critical Number A number c in the domain of f is a **critical number** if $f'(c) = 0$ or $f'(c)$ does not exist.

Closed Interval Method To find absolute extrema of a continuous function f on $[a, b]$:

1. Find all critical numbers in (a, b) .
2. Evaluate f at the critical numbers. Evaluate $f(a)$ and $f(b)$. Compare all values.

Theorem 1.2.3: Rolle's Theorem

Suppose f is continuous on $[a, b]$, f is differentiable on (a, b) , and $f(a) = f(b)$, then $\exists c \in (a, b)$ such that

$$f'(c) = 0$$

Theorem 1.2.4: Mean Value Theorem

Suppose f is continuous on $[a, b]$ and f is differentiable on (a, b) then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

First Derivative Test Let c be a critical number:

1. If f' changes from positive to negative at c , then f has a local maximum at c .
2. If f' changes from negative to positive at c , then f has a local minimum at c .
3. If f' does not change sign at c , then f has no local extremum at c .

Concavity

1. f is **concave up** on an interval if $f''(x) > 0$ on that interval.
2. f is **concave down** on an interval if $f''(x) < 0$ on that interval.

Second Derivative Test Suppose $f'(c) = 0$:

1. If $f''(c) > 0$, then f has a local minimum at c .
2. If $f''(c) < 0$, then f has a local maximum at c .
3. If $f''(c) = 0$, the test is inconclusive.

Inflection Point A point $x = c$ is an **inflection point** if f is continuous at c and concavity changes at c . Possible candidates occur when $f''(c) = 0$ or $f''(c)$ does not exist

Theorem 1.2.5: L'Hopital's Rule

If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ or both limits are infinite, and if $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists or equals $\pm\infty$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

1.3 Integrations and the Fundamental Theorem of Calculus

Area Approximation On $[a, b]$, $\Delta x = \frac{b-a}{n}$. Right endpoint: $x_i = a + i\Delta x$. Left endpoint: $x_{i-1} = a + (i-1)\Delta x$. Midpoint: $\bar{x}_i = a + (i - \frac{1}{2})\Delta x$. The approximate area is $\sum_{i=1}^n f(x_i^*)\Delta x$.

Definition 1.3.1: Definite Integral

If the limit exists, let $\Delta x = \frac{b-a}{n}$:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*)\Delta x$$

A definite integral is signed area. Area above the x-axis is positive; area below is negative.

Properties

$$\begin{aligned} \int_a^a f(x) dx &= 0 & \int_a^b f(x) dx &= -\int_b^a f(x) dx & \int_a^b cf(x) dx &= c \int_a^b f(x) dx \\ \int_a^b [f(x) + g(x)] dx &= \int_a^b f(x) dx + \int_a^b g(x) dx & \int_a^b f(x) dx &= \int_a^c f(x) dx + \int_c^b f(x) dx \end{aligned}$$

Antiderivative F is an antiderivative of f if $F'(x) = f(x)$. All antiderivatives are $F(x) + C$.

Theorem 1.3.2: Fundamental Theorem of Calculus Part 1

If f is continuous and $g(x) = \int_a^x f(t) dt$ then $g'(x) = f(x)$.

Theorem 1.3.3: Fundamental Theorem of Calculus Part 2

If f is continuous on $[a, b]$ and $F'(x) = f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

Substitution Rule If $u = g(x)$, $du = g'(x) dx$, then

$$\int f(g(x))g'(x) dx = \int f(u) du$$

For definite integrals, change the bounds:

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

Integration by Parts

$$\int u dv = uv - \int v du$$

For definite integrals,

$$\int_a^b u dv = [uv]_a^b - \int_a^b v du$$

Partial Fractions For a proper rational function $P(x)/Q(x)$ (where $\deg P < \deg Q$):

- **Distinct linear:** $(x - a)(x - b) \rightarrow \frac{A}{x-a} + \frac{B}{x-b}$
- **Repeated linear:** $(x - a)^r \rightarrow \frac{A_1}{x-a} + \cdots + \frac{A_r}{(x-a)^r}$
- **Irreducible quadratic:** $ax^2 + bx + c \rightarrow \frac{Ax+B}{ax^2+bx+c}$

Key integrals:

$$\int \frac{dx}{ax+b} = \frac{1}{a} \ln |ax+b| + C, \quad \int \frac{dx}{x^2+a^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

Infinite Intervals

$$\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$

$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$$

For $\int_{-\infty}^\infty f(x) dx$ split at any c :

$$\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^\infty f(x) dx.$$

Theorem 1.3.4: Comparison Theorem

If $0 \leq f(x) \leq g(x)$ on $[a, \infty)$, then:

1. If $\int_a^\infty g(x) dx$ converges, then $\int_a^\infty f(x) dx$ converges.
2. If $\int_a^\infty f(x) dx$ diverges, then $\int_a^\infty g(x) dx$ diverges.

1.4 Applications of Integration

Absolute Area If f crosses the x -axis, geometric area is below, split at zeros of f and remove negative signed area:

$$A = \int_a^b |f(x)| dx$$

Area Between Curves If $f(x) \geq g(x)$ on $[a, b]$, then

$$A = \int_a^b [f(x) - g(x)] dx$$

Surface Area If $y = f(x) \geq 0$ and f' is continuous on $[a, b]$, then rotating the curve around the x -axis gives

$$S = 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} dx$$

Chapter 2

Ordinary Differential Equations

2.1 Introduction to Differential Equations

2.1.1 Definitions and Terminology

Definition 2.1.1: Differential Equation

A *differential equation* is an equation that contains derivatives of one or more unknown functions with respect to one or more independent variables. The unknown function is usually called the *dependent variable*, and the variable with respect to which differentiation is taken is called the *independent variable*.

Ordinary and Partial Differential Equations A DE is called an *ordinary differential equation* (ODE) if it contains only ordinary derivatives with respect to a single independent variable. It is called a *partial differential equation* (PDE) if it contains partial derivatives of one or more unknown functions of two or more independent variables.

Example.

The equations $\frac{dy}{dx} + 5y = e^x$ and $\frac{d^2y}{dx^2} - 6y = 0$ are ordinary differential equations.
By contrast, $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$ is a partial differential equation.

Order. The *order* of a differential equation is the order of the highest derivative appearing in the equation. In one dependent variable, the general form of an n th-order ODE is

$$F(x, y, y', \dots, y^{(n)}) = 0$$

If the equation can be solved for the highest derivative, it can be written in *normal form*:

$$\frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n-1)})$$

Linear Ordinary Differential Equation An n th-order ODE is *linear* if it's linear in the dependent variable and all of its derivatives. Its general form is

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = g(x)$$

The coefficients may depend on the independent variable x , but not on y or its derivatives.

Example.

The equation $y'' - 2y' + y = e^x$ is linear.

The equations $yy' + x = 0$, $y'' + \sin y = 0$ and $(y')^2 + y = 0$ are nonlinear.

Solution of an ODE A function ϕ , defined on an interval I and possessing the required derivatives on I , is called a *solution* of an n th-order ODE if substituting $y = \phi(x)$ into the equation reduces it to an identity on I .

Interval of Definition The interval on which a solution is defined is called the *interval of definition*, *interval of existence*, *interval of validity*, or *domain of the solution*.

Explicit and Implicit Solutions An *explicit solution* expresses the dependent variable directly as a function of the independent variable: $y = \phi(x)$. An *implicit solution* is a relation $G(x, y) = 0$ that defines at least one differentiable function $y = \phi(x)$ satisfying the differential equation on an interval.

Example.

The relation $x^2 + y^2 = 25$ is an implicit solution of $\frac{dy}{dx} = -\frac{x}{y}$. On the interval $(-5, 5)$, it gives two explicit branches: $y = \sqrt{25 - x^2}$ and $y = -\sqrt{25 - x^2}$.

Families, Particular Solutions, and Singular Solutions A solution containing one or more constants is called a *family of solutions*. A solution obtained by assigning values to these constants is called a *particular solution*. A *singular solution* is a solution that can't be obtained from the family of solutions by any choice of the arbitrary constants.

Example.

The equation $y' = y$ has the one-parameter family of solutions $y = ce^x$. If the condition $y(0) = 3$ is imposed, then $c = 3$, so the particular solution is $y = 3e^x$.

System of ODEs A *system of ordinary differential equations* consists of two or more equations involving derivatives of two or more unknown functions of a single independent variable.

2.1.2 Initial-Value Problems**Definition 2.1.2: Initial-Value Problem**

An n th-order *initial-value problem* consists of an n th-order differential equation together with n side conditions imposed at the same point x_0 :

$$\frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n-1)})$$

subject to

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1}$$

Theorem 2.1.3: Existence and Uniqueness for a First-Order IVP

Let R be a rectangular region in the xy -plane containing the point (x_0, y_0) . If $f(x, y)$ and $\frac{\partial f}{\partial y}(x, y)$ are continuous on R , then there exists an interval $I_0 = (x_0 - h, x_0 + h)$ for some $h > 0$ on which the initial-value problem

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

has a unique solution.

This theorem is local: it guarantees a unique solution only near x_0 , not necessarily on the largest possible interval. It gives sufficient, not necessary, conditions. Roughly, f continuous gives existence, and f_y continuous gives uniqueness.

Example.

The initial-value problem $\frac{dy}{dx} = x\sqrt{y}$, $y(0) = 0$ has more than one solution. For example, $y = 0$ and $y = \frac{x^4}{16}$ both satisfy the differential equation and the initial condition. This does not contradict the existence–uniqueness theorem, because $\frac{\partial}{\partial y}(x\sqrt{y}) = \frac{x}{2\sqrt{y}}$ is not continuous at $y = 0$.

2.2 First-Order Differential Equations

2.2.1 Solution Curves Without an Explicit Solution

Direction Field Consider a first-order differential equation in normal form $\frac{dy}{dx} = f(x, y)$. At each point (x, y) where f is defined, the value $f(x, y)$ gives the slope of a small line segment, called a *lineal element*. A collection of such lineal elements drawn on a grid in the xy -plane is called a *direction field*.

Definition 2.2.1: Autonomous First-Order Differential Equation

A first-order differential equation is called *autonomous* if the independent variable does not appear explicitly. In normal form, it can be written as

$$\frac{dy}{dx} = f(y)$$

Critical Point and Equilibrium Solution A real number c is called a *critical point*, *equilibrium point*, or *stationary point* of $\frac{dy}{dx} = f(y)$ if $f(c) = 0$. In this case, the constant function $y(x) = c$ is a solution, called an *equilibrium solution*.

Phase Line and Phase Portrait For an autonomous equation $\frac{dy}{dx} = f(y)$ a *phase line* is a vertical line on which the critical points are marked. The sign of $f(y)$ on each interval between critical points determines whether solutions are increasing or decreasing. The resulting diagram is called a *one-dimensional phase portrait*.

Attractors, Repellers, and Semi-Stable Points Let c be a critical point of $\frac{dy}{dx} = f(y)$

- If nearby solutions approach c as $x \rightarrow \infty$, then c is an *attractor* or *asymptotically stable equilibrium*.
- If nearby solutions move away from c , then c is a *repeller* or *unstable equilibrium*.
- If solutions approach c from one side and move away from c from the other side, then c is *semi-stable*.

Proposition 2.2.2

If $y(x)$ is a solution of the autonomous equation $\frac{dy}{dx} = f(y)$ then for any constant k , $y_1(x) = y(x - k)$ is also a solution. This is called the *translation property*.

2.2.2 Separable Equations**Definition 2.2.3: Separable Equation**

A first-order differential equation of the form

$$\frac{dy}{dx} = g(x)h(y)$$

is called a *separable differential equation*.

Procedure. To solve a separable differential equation:

1. Write the equation in separable form $\frac{dy}{dx} = g(x)h(y)$.
2. Assuming $h(y) \neq 0$, separate the variables $\frac{1}{h(y)} dy = g(x) dx$.
3. Integrate both sides $\int \frac{1}{h(y)} dy = \int g(x) dx$.
4. Write the solution implicitly as $H(y) = G(x) + C$.

When solving a separable equation, any value r such that $h(r) = 0$ should be checked separately. The constant function $y = r$ may be a solution, but it may be lost after division by $h(y)$.

2.2.3 Linear Equations**Definition 2.2.4: First-Order Linear Equation**

A first-order differential equation of the form $a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$ is called a *linear differential equation* in the dependent variable y . If $a_1(x) \neq 0$, division by $a_1(x)$ gives the standard form

$$\frac{dy}{dx} + P(x)y = f(x)$$

Procedure. To solve a first-order linear equation:

1. Put the equation in standard form $\frac{dy}{dx} + P(x)y = f(x)$.
2. Compute the integrating factor $\mu(x) = e^{\int P(x) dx}$.
3. Multiply both sides by $\mu(x)$ and rewrite the left-hand side as $\frac{d}{dx}(\mu(x)y)$.
4. Integrate both sides and solve for y .

$$y = e^{-\int P(x) dx} \left(\int e^{\int P(x) dx} f(x) dx + C \right)$$

2.2.4 Exact Differential Equations

Definition 2.2.5: Exact Differential Equation

A first-order differential equation written in differential form

$$M(x, y) dx + N(x, y) dy = 0$$

is called an *exact equation* if there exists a function $\Phi(x, y)$ such that

$$\frac{\partial \Phi}{\partial x} = M(x, y), \quad \frac{\partial \Phi}{\partial y} = N(x, y)$$

In that case, $d\Phi = M dx + N dy$ and the solution is given implicitly by $\Phi(x, y) = C$.

Theorem 2.2.6: Criterion for Exactness

Suppose $M(x, y)$ and $N(x, y)$ have continuous first partial derivatives in a rectangular region R . Then $M(x, y) dx + N(x, y) dy$ is an exact differential in R if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Procedure.

1. Integrate M with respect to x : $\Phi(x, y) = \int M(x, y) dx + g(y)$.
2. Differentiate this expression with respect to y : $\Phi_y = \frac{\partial}{\partial y} \left(\int M(x, y) dx \right) + g'(y)$.
3. Set $\Phi_y = N(x, y)$ and solve for $g'(y)$. Integrate to find $g(y)$.
4. Write the solution as $\Phi(x, y) = C$.

Integrating Factors for Nonexact Equations. If $M(x, y) dx + N(x, y) dy = 0$ is not exact, it may become exact after multiplication by a nonzero function $\mu(x, y)$: $\mu M dx + \mu N dy = 0$. Two useful special cases are:

$$\begin{cases} \mu(x) = \exp \left(\int \frac{M_y - N_x}{N} dx \right), & \text{if } \frac{M_y - N_x}{N} \text{ depends only on } x \\ \mu(y) = \exp \left(\int \frac{N_x - M_y}{M} dy \right), & \text{if } \frac{N_x - M_y}{M} \text{ depends only on } y \end{cases}$$

2.2.5 Solutions by Substitutions

Definition 2.2.7: Substitution Method

For a first-order equation $\frac{dy}{dx} = f(x, y)$ introduce a substitution $y = g(x, u)$, $u = u(x)$. By the chain rule,

$$\frac{dy}{dx} = \frac{\partial g}{\partial x}(x, u) + \frac{\partial g}{\partial u}(x, u) \frac{du}{dx}$$

After substitution, the original equation may become a simpler equation $\frac{du}{dx} = F(x, u)$. If this new equation is separable or linear, it can be solved by earlier methods. Finally, substitute back to obtain y .

Solving a Homogeneous Equation. If the equation is homogeneous, use the substitution

$$y = ux, \quad \frac{dy}{dx} = u + x \frac{du}{dx}$$

Solving a Bernoulli Equation. A differential equation of the form $\frac{dy}{dx} + P(x)y = f(x)y^n$ is called a *Bernoulli equation*. For $n \neq 0, 1$, it can be reduced to a linear equation by the substitution

$$u = y^{1-n}, \quad \frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx}$$

Reduction to Separation of Variables A differential equation of the form $\frac{dy}{dx} = f(Ax + By + C)$, $B \neq 0$ can be reduced to a separable equation by the substitution

$$u = Ax + By + C, \quad \frac{du}{dx} = A + B \frac{dy}{dx}$$

2.2.6 A Numerical Method

Definition 2.2.8: Euler's Method

Consider the initial-value problem $y' = f(x, y)$, $y(x_0) = y_0$, Euler's method approximates the solution by repeatedly following tangent lines:

$$x_n = x_0 + nh, \quad y_{n+1} = y_n + hf(x_n, y_n), \quad n = 0, 1, 2, \dots$$

2.3 Higher-Order Differential Equations

2.3.1 Theory of Linear Equations

Linear n th-Order Differential Equation A linear n th-order differential equation has the form

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = g(x)$$

where $a_n(x) \neq 0$ on an interval I . If $g(x) = 0$, the equation is *homogeneous*; otherwise it is *nonhomogeneous*. An n th-order initial-value problem specifies the value of y and its first $n - 1$ derivatives at the same point x_0 :

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1}$$

Theorem 2.3.1: Existence and Uniqueness for Linear IVPs

Suppose $a_n, a_{n-1}, \dots, a_0$ and g are continuous on an interval I , and $a_n(x) \neq 0$ for every $x \in I$. Then for any $x_0 \in I$, the corresponding n th-order linear initial-value problem has a unique solution on I .

This theorem is much stronger than the first-order local theorem: for linear equations, the solution exists on the whole interval on which the coefficient functions are continuous and the leading coefficient is nonzero.

Boundary-Value Problems. A boundary-value problem imposes conditions at different points, for example $y(a) = y_0, y(b) = y_1$. Unlike initial-value problems, a boundary-value problem may have infinitely many solutions, exactly one solution, or no solution.

Differential Operator Notation. Let $D = \frac{d}{dx}$. Then $D^n y = \frac{d^n y}{dx^n}$. The linear differential operator is

$$L = a_n(x)D^n + a_{n-1}(x)D^{n-1} + \cdots + a_1(x)D + a_0(x)$$

so the differential equation can be written compactly as $L[y] = g(x)$.

Proposition 2.3.2

The operator L is linear, where α and β are constants:

$$L[\alpha f + \beta g] = \alpha L[f] + \beta L[g],$$

Theorem 2.3.3: Superposition Principle for Homogeneous Equations

If $y_1, \dots, y_k$ are solutions of the homogeneous equation $L[y] = 0$ on an interval I , then every linear combination $y = c_1 y_1 + \cdots + c_k y_k$ is also a solution on I .

Definition 2.3.4: Linear Independence and the Wronskian

Functions $f_1, \dots, f_n$ are linearly dependent on I if there exist constants $c_1, \dots, c_n$, not all zero, such that

$$c_1 f_1(x) + \cdots + c_n f_n(x) = 0$$

for every $x \in I$. Otherwise they are linearly independent. The *Wronskian* of $f_1, \dots, f_n$ is

$$W(f_1, \dots, f_n) = \begin{vmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix}$$

Wronskian Criterion for Solutions wronskiancriterion Let $y_1, \dots, y_n$ be n solutions of a homogeneous linear n th-order differential equation on I . Then they are linearly independent on I if and only if for every $x \in I$:

$$W(y_1, \dots, y_n)(x) \neq 0$$

Fundamental Set of Solutions A set of n linearly independent solutions of a homogeneous linear n th-order differential equation is called a *fundamental set of solutions*.

Theorem 2.3.5: General Solution of a Homogeneous Linear Equation

If $y_1, \dots, y_n$ form a fundamental set of solutions for $L[y] = 0$ on I , then the general solution is

$$y = c_1 y_1 + \cdots + c_n y_n$$

Theorem 2.3.6: General Solution of a Nonhomogeneous Linear Equation

If y_p is one particular solution of $L[y] = g(x)$ and $y_c = c_1y_1 + \cdots + c_ny_n$ is the general solution of the associated homogeneous equation $L[y] = 0$, then the general solution of the nonhomogeneous equation is

$$y = y_c + y_p$$

Proposition 2.3.7

If $y_{p1}, \dots, y_{pk}$ are particular solutions of

$$L[y] = g_1(x), \quad \dots, \quad L[y] = g_k(x)$$

then $y_p = y_{p1} + \cdots + y_{pk}$ is a particular solution of

$$L[y] = g_1(x) + \cdots + g_k(x)$$

2.3.2 Reduction of Order**Theorem 2.3.8: Reduction of Order Formula**

Let $y_1(x)$ be a nonzero solution of

$$y'' + P(x)y' + Q(x)y = 0$$

on an interval I . Then a second solution is

$$y_2(x) = y_1(x) \int \frac{e^{-\int P(x) dx}}{[y_1(x)]^2} dx$$

Procedure.

1. Put the equation in standard form $y'' + P(x)y' + Q(x)y = 0$.
2. Identify a known nonzero solution y_1 .
3. Use $y_2 = y_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx$.
4. Write the general solution as $y = c_1y_1 + c_2y_2$.

2.3.3 Homogeneous Linear Equations with Constant Coefficients

Auxiliary Equation. For $a_ny^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1y' + a_0y = 0$, try $y = e^{mx}$. Substitution gives the *auxiliary equation* $a_n m^n + a_{n-1} m^{n-1} + \cdots + a_1 m + a_0 = 0$. The roots of this polynomial determine the solution.

Second-Order Case. For $ay'' + by' + cy = 0$ the auxiliary equation is $am^2 + bm + c = 0$.

Root type	General solution
$m_1 \neq m_2$ real	$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$
$m_1 = m_2 = m$	$y = c_1 e^{mx} + c_2 x e^{mx}$
$m = \alpha \pm i\beta$	$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$

Higher-Order Roots. If m is a real root of multiplicity k , then the corresponding solutions are

$$e^{mx}, \quad xe^{mx}, \quad x^2e^{mx}, \quad \dots, \quad x^{k-1}e^{mx}$$

If $\alpha \pm i\beta$ is a complex conjugate pair of multiplicity k , then the real solutions are

$$e^{\alpha x} \cos \beta x, \quad xe^{\alpha x} \cos \beta x, \quad \dots, \quad x^{k-1}e^{\alpha x} \cos \beta x$$

$$e^{\alpha x} \sin \beta x, \quad xe^{\alpha x} \sin \beta x, \quad \dots, \quad x^{k-1}e^{\alpha x} \sin \beta x$$

2.3.4 Undetermined Coefficients: Superposition Approach

When the Method Applies. The method of undetermined coefficients applies to nonhomogeneous linear differential equations with constant coefficients, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$, when $g(x)$ is a finite sum of constants, polynomials, exponentials, sines, cosines, or products of these functions.

Procedure.

1. Solve the associated homogeneous equation to find y_c .
2. Guess the form of a particular solution y_p based on $g(x)$.
3. Substitute y_p into the original equation and solve for the unknown coefficients.
4. Write the general solution as $y = y_c + y_p$.

Common Trial Forms.

$g(x)$	Trial form for y_p
k	A
$a_m x^m + \dots + a_0$	$A_m x^m + \dots + A_0$
$e^{\alpha x}$	$Ae^{\alpha x}$
$x^m e^{\alpha x}$	$(A_m x^m + \dots + A_0)e^{\alpha x}$
$\cos \beta x$ or $\sin \beta x$	$A \cos \beta x + B \sin \beta x$
$e^{\alpha x} \cos \beta x$ or $e^{\alpha x} \sin \beta x$	$e^{\alpha x}(A \cos \beta x + B \sin \beta x)$

If any term in the guessed y_p is already contained in y_c , multiply the whole guess by the smallest power x^s that removes the duplication: $y_p = x^s(\text{original guess})$.

2.3.5 Undetermined Coefficients: Annihilator Approach

Definition 2.3.9: Annihilator

A differential operator L with constant coefficients is called an *annihilator* of $f(x)$ if

$$L[f(x)] = 0$$

Basic Annihilators.

$$D^n \text{ annihilates } 1, x, x^2, \dots, x^{n-1}.$$

$$(D - \alpha)^n \text{ annihilates } e^{\alpha x}, xe^{\alpha x}, \dots, x^{n-1}e^{\alpha x}.$$

$$[(D - \alpha)^2 + \beta^2]^n \text{ annihilates } x^k e^{\alpha x} \cos \beta x \text{ and } x^k e^{\alpha x} \sin \beta x, k = 0, 1, \dots, n-1.$$

Procedure. To solve $L[y] = g(x)$ by annihilators:

1. Solve $L[y] = 0$ to find the complementary function y_c .
2. Find an operator L_1 such that $L_1[g] = 0$.
3. Apply L_1 to obtain the homogeneous equation $L_1L[y] = 0$.
4. Solve the higher-order homogeneous equation.
5. Remove the terms already in y_c ; the remaining terms give the form of y_p .
6. Substitute into the original equation to determine the coefficients.

The annihilator method justifies the guessing method. It is often more useful for finding the correct form of y_p than for doing the final coefficient calculation.

2.3.6 Variation of Parameters

Procedure.

1. Put the equation in standard form $y'' + P(x)y' + Q(x)y = f(x)$.
2. Solve the homogeneous equation to find y_1, y_2 .
3. Compute $W = y_1y_2' - y_2y_1'$. Compute u_1' and u_2' using $u_1' = -\frac{y_2f}{W}, u_2' = \frac{y_1f}{W}$.
4. Integrate to find u_1, u_2 , then write $y_p = u_1y_1 + u_2y_2$.
5. The general solution is $y = y_c + y_p$.

2.3.7 Cauchy–Euler Equations

Definition 2.3.10: Cauchy–Euler Equation

A second-order Cauchy–Euler equation has the form where a, b, c are constants:

$$ax^2y'' + bxy' + cy = g(x)$$

More generally, an n th-order Cauchy–Euler equation has powers of x matched to the order of the derivative:

$$a_nx^ny^{(n)} + a_{n-1}x^{n-1}y^{(n-1)} + \cdots + a_1xy' + a_0y = g(x)$$

Root Cases. On an interval where $x > 0$:

Root type	General solution
$m_1 \neq m_2$ real	$y = c_1x^{m_1} + c_2x^{m_2}$
$m_1 = m_2 = m$	$y = c_1x^m + c_2x^m \ln x$
$m = \alpha \pm i\beta$	$y = x^\alpha (c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x))$

Nonhomogeneous Case.

1. Solve the associated homogeneous Cauchy–Euler equation.
2. Divide by the leading coefficient to put the equation in standard form.
3. Use variation of parameters to find y_p .

2.3.8 Solving Systems of Linear Differential Equations by Elimination

Definition 2.3.11: Solution of a System

A solution of a system of differential equations is a set of sufficiently differentiable functions, such as

$$x = \Phi_1(t), \quad y = \Phi_2(t), \quad z = \Phi_3(t)$$

that satisfies every equation in the system on a common interval I .

Procedure.

1. Rewrite the system using the differential operator D .
2. Apply operators to the equations so that one unknown can be eliminated.
3. Solve the resulting higher-order equation for one dependent variable.
4. Substitute back into one original equation to find the other variables.
5. Use the original system to relate the arbitrary constants.

2.4 Series Solutions of Linear Equations

2.4.1 Review of Power Series

Definition 2.4.1: Power Series

A *power series centered at a* is an infinite series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots$$

Convergence and Radius of Convergence. A power series may converge for some values of x and diverge for others. The set of all real numbers x for which the series converges is called its *interval of convergence*. There is a number R , called the *radius of convergence*, such that $\sum_{n=0}^{\infty} c_n(x-a)^n$ converges for $|x-a| < R$ and diverges for $|x-a| > R$. At the endpoints $x = a \pm R$, convergence must be checked separately.

Ratio Test. For a power series, the ratio test often gives the radius of convergence. If $L = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n} \right|$ then the series converges absolutely when $L < 1$ and diverges when $L > 1$. The test is inconclusive when $L = 1$, which usually corresponds to the endpoints.

Definition 2.4.2: Analytic Function

A function f is *analytic at a* if it can be represented by a power series centered at a with $R > 0$:

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$$

Proposition 2.4.3

If $\sum_{n=0}^{\infty} c_n(x-a)^n = 0$ for all x in some open interval around a , then every coefficient must be zero: $c_n = 0, n = 0, 1, 2, \dots$. This is called the *identity property* of power series.

2.4.2 Solutions About Ordinary Points**Definition 2.4.4: Ordinary and Singular Points**

Consider the homogeneous linear second-order equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$. After division by $a_2(x)$, write it in standard form $y'' + P(x)y' + Q(x)y = 0$. A point $x = x_0$ is an *ordinary point* if both $P(x)$ and $Q(x)$ are analytic at x_0 . Otherwise, $x = x_0$ is a *singular point*.

Theorem 2.4.5: Existence of Power Series Solutions

If $x = x_0$ is an ordinary point of $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ then the equation has two linearly independent solutions of the form $y = \sum_{n=0}^{\infty} c_n(x-x_0)^n$. Each power series solution converges at least for $|x-x_0| < R$ where R is the distance from x_0 to the nearest singular point.

Procedure. To find a power series solution about an ordinary point:

1. Assume $y = \sum_{n=0}^{\infty} c_n(x-x_0)^n$.
2. Differentiate term by term to find y' and y'' . Substitute y, y', y'' into the differential equation. Make all series use the same power of $x-x_0$. Set each coefficient equal to zero. Solve the resulting recurrence relation.
3. Write the solution in terms of the arbitrary constants, usually c_0 and c_1 .

2.4.3 Solutions About Singular Points**Definition 2.4.6: Regular and Irregular Singular Points**

Suppose $x = x_0$ is a singular point of $y'' + P(x)y' + Q(x)y = 0$. Then $x = x_0$ is a *regular singular point* if

$$p(x) = (x-x_0)P(x), \quad q(x) = (x-x_0)^2Q(x)$$

are both analytic at x_0 . A singular point that is not regular is called an *irregular singular point*.

Theorem 2.4.7: Frobenius' Theorem

If $x = x_0$ is a regular singular point of $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$, then there exists at least one solution of the following form where r is a constant determined by the differential equation

$$y = \sum_{n=0}^{\infty} c_n(x-x_0)^{n+r} = (x-x_0)^r \sum_{n=0}^{\infty} c_n(x-x_0)^n, \quad c_0 \neq 0$$

Indicial Equation. For a regular singular point at $x = 0$, write $y'' + P(x)y' + Q(x)y = 0$, and define

$$p(x) = xP(x) = a_0 + a_1x + a_2x^2 + \dots, \quad q(x) = x^2Q(x) = b_0 + b_1x + b_2x^2 + \dots$$

Then the *indicial equation* is $r(r-1) + a_0r + b_0 = 0$. Its roots are called the *indicial roots*.

Procedure. To solve near a regular singular point $x = 0$:

1. Assume a Frobenius solution $y = \sum_{n=0}^{\infty} c_n x^{n+r}$, $c_0 \neq 0$.
2. Compute y' and y'' . Substitute into the differential equation. Set the coefficient of the lowest power of x equal to zero. This gives the indicial equation. Solve the recurrence relation for each indicial root.
3. Determine whether one or two linearly independent Frobenius series are obtained.

Three Cases for Indicial Roots. Assume r_1 and r_2 are real indicial roots with $r_1 \geq r_2$.

Case	Form of solutions
$r_1 - r_2 \notin \mathbb{Z}_{>0}$	Two Frobenius series usually exist
$r_1 - r_2 \in \mathbb{Z}_{>0}$	One Frobenius series always exists; the second may contain $\ln x$
$r_1 = r_2$	Second solution contains $\ln x$

More explicitly $y_1 = \sum_{n=0}^{\infty} c_n x^{n+r_1}$, $c_0 \neq 0$. The second solution may have the form

$$y_2 = Cy_1 \ln x + \sum_{n=0}^{\infty} b_n x^{n+r_2}, \quad b_0 \neq 0,$$

where C may be zero in the second case, but is nonzero in the repeated-root case.

2.5 The Laplace Transform

2.5.1 Definition of the Laplace Transform

Definition 2.5.1: Laplace Transform

Let f be defined for $t \geq 0$. The *Laplace transform* of f is

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt = \lim_{b \rightarrow \infty} \int_0^b e^{-st} f(t) dt$$

The variable t is the original variable, while s is the transform variable.

Basic Transform Table. For suitable values of s ,

$f(t)$	$\mathcal{L}\{f(t)\} = F(s)$
1	$\frac{1}{s}$
t^n	$\frac{n!}{s^{n+1}}, \quad n = 1, 2, 3, \dots$
e^{at}	$\frac{1}{s-a}$
$\sin kt$	$\frac{k}{s^2 + k^2}$
$\cos kt$	$\frac{s}{s^2 + k^2}$
$\sinh kt$	$\frac{k}{s^2 - k^2}$
$\cosh kt$	$\frac{s}{s^2 - k^2}$

Proposition 2.5.2

The Laplace transform is linear. If $\mathcal{L}\{f\} = F$ and $\mathcal{L}\{g\} = G$, then

$$\mathcal{L}\{\alpha f(t) + \beta g(t)\} = \alpha F(s) + \beta G(s)$$

Definition 2.5.3: Piecewise Continuity and Exponential Order

A function f is *piecewise continuous* on $[0, \infty)$ if, on every finite interval, it is continuous except possibly at finitely many points, and each discontinuity is finite. A function f is of *exponential order* c as $t \rightarrow \infty$ if there exist constants $M > 0$, $c > 0$, and $T > 0$ such that

$$|f(t)| \leq Me^{ct}, \quad t > T$$

Theorem 2.5.4: Existence of the Laplace Transform

If f is piecewise continuous on $[0, \infty)$ and is of exponential order c , then $\mathcal{L}\{f(t)\}$ exists for $s > c$.

2.5.2 Inverse Transforms and Transforms of Derivatives**Definition 2.5.5: Inverse Laplace Transform**

If $\mathcal{L}\{f(t)\} = F(s)$, then $f(t)$ is called the *inverse Laplace transform* of $F(s)$, written

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

Partial Fractions. For rational functions, first decompose into simpler fractions. Typical patterns are

$$\frac{P(s)}{(s-a)(s-b)} = \frac{A}{s-a} + \frac{B}{s-b}$$

$$\frac{P(s)}{(s-a)^m} = \frac{A_1}{s-a} + \frac{A_2}{(s-a)^2} + \cdots + \frac{A_m}{(s-a)^m}$$

Theorem 2.5.6: Transform of a Derivative

Suppose the needed derivatives of f are continuous or piecewise continuous and of exponential order. If $\mathcal{L}\{f(t)\} = F(s)$, then

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$$

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0), \quad \mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$

Solving Linear IVPs by Laplace Transform. For a constant-coefficient linear initial-value problem:

1. Take the Laplace transform of both sides. Use initial conditions through the derivative formulas.
2. Solve the algebraic equation for $Y(s)$. Decompose $Y(s)$ into recognizable pieces. Apply $\mathcal{L}^{-1}$ to obtain $y(t)$.

2.5.3 Operational Properties

Theorem 2.5.7: First Translation Theorem

If $\mathcal{L}\{f(t)\} = F(s)$, then

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a) \Leftrightarrow \mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$$

Definition 2.5.8: Unit Step Function

The *unit step function* at a is

$$U(t-a) = \begin{cases} 0, & 0 \leq t < a, \\ 1, & t \geq a \end{cases}$$

Theorem 2.5.9: Second Translation Theorem

If $\mathcal{L}\{f(t)\} = F(s)$ and $a > 0$, then

$$\mathcal{L}\{U(t-a)f(t-a)\} = e^{-as}F(s) \Leftrightarrow \mathcal{L}^{-1}\{e^{-as}F(s)\} = U(t-a)f(t-a)$$

Alternative Form. For a term of the form $g(t)U(t-a)$, it is often easier to use

$$\mathcal{L}\{g(t)U(t-a)\} = e^{-as}\mathcal{L}\{g(t+a)\}$$

Theorem 2.5.10: Derivatives of Transforms

If $\mathcal{L}\{f(t)\} = F(s)$, then for $n = 1, 2, 3, \dots$,

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

$$\mathcal{L}\{tf(t)\} = -F'(s)$$

Definition 2.5.11: Convolution

If f and g are piecewise continuous on $[0, \infty)$, their *convolution* is

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau) d\tau$$

Basic Properties of Convolution. Convolution behaves like a generalized product:

$$f * g = g * f, \quad (f * g) * h = f * (g * h), \quad f * (g + h) = f * g + f * h$$

Theorem 2.5.12: Convolution Theorem

If $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{g(t)\} = G(s)$, then

$$\mathcal{L}\{(f * g)(t)\} = F(s)G(s) \Leftrightarrow \mathcal{L}^{-1}\{F(s)G(s)\} = f * g$$

Transform of an Integral. Taking $g(t) = 1$ in the convolution theorem gives

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}, \quad F(s) = \mathcal{L}\{f(t)\}$$

Definition 2.5.13: Volterra Integral Equation

An equation of the following form is a *Volterra integral equation*. The integral term is a convolution, so Laplace transforms turn the equation into an algebraic equation for $F(s) = \mathcal{L}\{f(t)\}$

$$f(t) = g(t) + \int_0^t f(\tau)h(t-\tau) d\tau$$

Theorem 2.5.14: Transform of a Periodic Function

If f is piecewise continuous on $[0, \infty)$ and periodic with period T , then

$$\mathcal{L}\{f(t)\} = \frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}$$

2.5.4 The Dirac Delta Function

Unit Impulse. A unit impulse models a force of very large magnitude acting over a very short time. A rectangular impulse centered near t_0 has total area 1, and as its width tends to zero, it is idealized by the Dirac delta.

Definition 2.5.15: Dirac Delta

The *Dirac delta* $\delta(t - t_0)$ is not an ordinary function. It is a generalized function characterized formally by

$$\delta(t - t_0) = 0 \quad (t \neq t_0), \quad \int_0^\infty \delta(t - t_0) dt = 1$$

Theorem 2.5.16: Transform of the Dirac Delta Function

For $t_0 > 0$,

$$\mathcal{L}\{\delta(t - t_0)\} = e^{-st_0}$$

2.6 Systems of Linear Differential Equations

2.6.1 Theory of Linear Systems

Definition 2.6.1: First-Order System

A system of n first-order differential equations has normal form

$$\frac{dx_i}{dt} = g_i(t, x_1, x_2, \dots, x_n), \quad i = 1, 2, \dots, n$$

If each g_i is linear in the dependent variables, the system is linear.

Matrix Form. If $\mathbf{F}(t) = \mathbf{0}$, the system is *homogeneous*; otherwise it's *nonhomogeneous*.

$$\mathbf{X}' = A(t)\mathbf{X} + \mathbf{F}(t), \quad \mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{F}(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{pmatrix}$$

Any n th-order equation $y^{(n)} = f(t, y, y', \dots, y^{(n-1)})$ can be rewritten as a first-order system by setting $x_1 = y, x_2 = y', \dots, x_n = y^{(n-1)}$. Then $x_1' = x_2, x_2' = x_3, \dots, x_{n-1}' = x_n, x_n' = f(t, x_1, x_2, \dots, x_n)$

Definition 2.6.2: Solution Vector

A *solution vector* on an interval I is a column vector whose entries are differentiable and satisfy the system on I .

$$\mathbf{X}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$$

Theorem 2.6.3: Existence and Uniqueness for Linear Systems

Suppose the entries of $A(t)$ and $\mathbf{F}(t)$ are continuous on an interval I containing t_0 . Then the initial-value problem $\mathbf{X}' = A(t)\mathbf{X} + \mathbf{F}(t), \mathbf{X}(t_0) = \mathbf{X}_0$ has a unique solution on I .

Theorem 2.6.4: Superposition for Homogeneous Systems

If $\mathbf{X}_1, \dots, \mathbf{X}_k$ are solution vectors of $\mathbf{X}' = A(t)\mathbf{X}$ on I , then $\mathbf{X} = c_1\mathbf{X}_1 + \dots + c_k\mathbf{X}_k$ is also a solution.

Definition 2.6.5: Linear Independence of Solution Vectors

Solution vectors $\mathbf{X}_1, \dots, \mathbf{X}_k$ are linearly dependent on I if there exist constants $c_1, \dots, c_k$, not all zero, such that

$$c_1\mathbf{X}_1(t) + \dots + c_k\mathbf{X}_k(t) = \mathbf{0}$$

for every $t \in I$. Otherwise, they are linearly independent.

Fundamental Matrix. For n linearly independent solutions $\mathbf{X}_1, \dots, \mathbf{X}_n$ (fundamental set of solutions), the matrix

$$\Phi(t) = \begin{pmatrix} | & | & & | \\ \mathbf{X}_1 & \mathbf{X}_2 & \cdots & \mathbf{X}_n \\ | & | & & | \end{pmatrix}$$

is called a *fundamental matrix*. Its determinant is the Wronskian of the solution vectors. the set of solution vectors is linearly independent on I if and only if the Wronskian $W(X_1, X_2, \dots, X_n) = |\Phi(t)| \neq 0$ for every t in the interval.

Theorem 2.6.6: General Solution of a Homogeneous Linear System

If $\Phi(t)$ is a fundamental matrix for $\mathbf{X}' = A(t)\mathbf{X}$, then the general solution is $\mathbf{X}(t) = \Phi(t)\mathbf{C}$ where $\mathbf{C}$ is a constant column vector.

Theorem 2.6.7: General Solution of a Nonhomogeneous System

Let $\mathbf{X}_p$ be a particular solution of $\mathbf{X}' = A(t)\mathbf{X} + \mathbf{F}(t)$, and let $\mathbf{X}_c = \Phi(t)\mathbf{C}$ be the general solution of the associated homogeneous system $\mathbf{X}' = A(t)\mathbf{X}$. Then the general solution is $\mathbf{X}(t) = \mathbf{X}_c(t) + \mathbf{X}_p(t)$.

2.6.2 Homogeneous Linear Systems

Procedure To solve $\mathbf{X}' = A\mathbf{X}$ use the following steps:

1. Find the eigenvalues by solving $\det(A - \lambda I) = 0$.
2. For each eigenvalue λ , find eigenvectors by solving $(A - \lambda I)\mathbf{K} = \mathbf{0}$.
3. If A has n linearly independent eigenvectors, form n solutions $\mathbf{K}_i e^{\lambda_i t}$.
4. If an eigenvalue is repeated but has too few eigenvectors, find generalized eigenvectors. If eigenvalues are complex, use the real and imaginary parts of $\mathbf{K}e^{\lambda t}$ to form real-valued solutions.

Distinct Real Eigenvalues distinct real eigenvalues If A has n distinct real eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ with corresponding eigenvectors $\mathbf{K}_1, \mathbf{K}_2, \dots, \mathbf{K}_n$, then the general solution of $\mathbf{X}' = A\mathbf{X}$ is

$$\mathbf{X}(t) = c_1 \mathbf{K}_1 e^{\lambda_1 t} + c_2 \mathbf{K}_2 e^{\lambda_2 t} + \dots + c_n \mathbf{K}_n e^{\lambda_n t}$$

Repeated Eigenvalues: Enough Eigenvectors. Suppose λ is an eigenvalue with algebraic multiplicity m . If there are m linearly independent eigenvectors $\mathbf{K}_1, \mathbf{K}_2, \dots, \mathbf{K}_m$, then the corresponding part of the general solution is

$$c_1 \mathbf{K}_1 e^{\lambda t} + c_2 \mathbf{K}_2 e^{\lambda t} + \dots + c_m \mathbf{K}_m e^{\lambda t}$$

Multiplicity Two with One Eigenvector. Suppose λ has algebraic multiplicity 2 but only one eigenvector $\mathbf{K}$. The first solution is $\mathbf{X}_1 = \mathbf{K}e^{\lambda t}$. To find a second solution, solve for a generalized eigenvector $\mathbf{P}$: $(A - \lambda I)\mathbf{P} = \mathbf{K}$. Then a second linearly independent solution is $\mathbf{X}_2 = (t\mathbf{K} + \mathbf{P})e^{\lambda t}$.

Multiplicity Three with One Eigenvector. Suppose λ has algebraic multiplicity 3 but only one eigenvector $\mathbf{K}$. First solve $(A - \lambda I)\mathbf{K} = \mathbf{0}$. Then find generalized eigenvectors $\mathbf{P}$ and $\mathbf{Q}$ satisfying $(A - \lambda I)\mathbf{P} = \mathbf{K}$, $(A - \lambda I)\mathbf{Q} = \mathbf{P}$. The three linearly independent solutions are

$$\begin{cases} \mathbf{X}_1 = \mathbf{K}e^{\lambda t} \\ \mathbf{X}_2 = (t\mathbf{K} + \mathbf{P})e^{\lambda t} \\ \mathbf{X}_3 = \left(\frac{t^2}{2}\mathbf{K} + t\mathbf{P} + \mathbf{Q}\right)e^{\lambda t} \end{cases}$$

General Repeated Eigenvalue with One Eigenvector Suppose λ has algebraic multiplicity m but only one eigenvector. We first find a generalized eigenvector chain $(A - \lambda I)\mathbf{K}_1 = \mathbf{0}, (A - \lambda I)\mathbf{K}_2 = \mathbf{K}_1, (A - \lambda I)\mathbf{K}_3 = \mathbf{K}_2, \dots, (A - \lambda I)\mathbf{K}_m = \mathbf{K}_{m-1}$. Here $\mathbf{K}_1$ is the ordinary eigenvector, while $\mathbf{K}_2, \dots, \mathbf{K}_m$ are generalized eigenvectors.

Then m linearly independent solutions are

$$\begin{cases} \mathbf{X}_1 = \mathbf{K}_1 e^{\lambda t} \\ \mathbf{X}_2 = (t\mathbf{K}_1 + \mathbf{K}_2) e^{\lambda t} \\ \mathbf{X}_3 = \left(\frac{t^2}{2!} \mathbf{K}_1 + t\mathbf{K}_2 + \mathbf{K}_3 \right) e^{\lambda t} \\ \dots \\ \mathbf{X}_j = \left(\frac{t^{j-1}}{(j-1)!} \mathbf{K}_1 + \frac{t^{j-2}}{(j-2)!} \mathbf{K}_2 + \dots + t\mathbf{K}_{j-1} + \mathbf{K}_j \right) e^{\lambda t}, \quad j = 1, 2, \dots, m \end{cases}$$

Complex Eigenvalues. Suppose A has real entries and has a complex eigenvalue $\lambda = \alpha + i\beta, \beta > 0$. The conjugate $\bar{\lambda} = \alpha - i\beta$ is also an eigenvalue. Let the eigenvector corresponding to $\lambda = \alpha + i\beta$ be $\mathbf{K} = \mathbf{B} + i\mathbf{C}$, where $\mathbf{B}$ is the real part of $\mathbf{K}$ and $\mathbf{C}$ is the imaginary part of $\mathbf{K}$. Then two real linearly independent solutions are

$$\begin{cases} \mathbf{X}_1 = e^{\alpha t} (\mathbf{B} \cos \beta t - \mathbf{C} \sin \beta t) \\ \mathbf{X}_2 = e^{\alpha t} (\mathbf{B} \sin \beta t + \mathbf{C} \cos \beta t) \end{cases}$$

Summary of Cases. For $\mathbf{X}' = A\mathbf{X}$ the form of the general solution:

Case	Solution Form
Distinct real eigenvalues	$\mathbf{K}_i e^{\lambda_i t}$
Repeated eigenvalue with enough eigenvectors	$\mathbf{K}_1 e^{\lambda t}, \dots, \mathbf{K}_m e^{\lambda t}$
Multiplicity two with one eigenvector	$\mathbf{K} e^{\lambda t}, (t\mathbf{K} + \mathbf{P}) e^{\lambda t}$
Multiplicity three with one eigenvector	$\mathbf{K} e^{\lambda t}, (t\mathbf{K} + \mathbf{P}) e^{\lambda t}, \left(\frac{t^2}{2} \mathbf{K} + t\mathbf{P} + \mathbf{Q} \right) e^{\lambda t}$
General repeated eigenvalue with one eigenvector	$\mathbf{X}_j = \left(\frac{t^{j-1}}{(j-1)!} \mathbf{K}_1 + \frac{t^{j-2}}{(j-2)!} \mathbf{K}_2 + \dots + t\mathbf{K}_{j-1} + \mathbf{K}_j \right) e^{\lambda t}$
Complex eigenvalues $\alpha \pm i\beta$	$e^{\alpha t} (\mathbf{B} \cos \beta t - \mathbf{C} \sin \beta t), e^{\alpha t} (\mathbf{B} \sin \beta t + \mathbf{C} \cos \beta t)$

2.6.3 Nonhomogeneous Linear Systems

General Form. A nonhomogeneous linear system has the form $\mathbf{X}' = A\mathbf{X} + \mathbf{F}(t)$. Its general solution is $\mathbf{X} = \mathbf{X}_c + \mathbf{X}_p$ where $\mathbf{X}_c$ solves the associated homogeneous system and $\mathbf{X}_p$ is one particular solution.

Undetermined Coefficients. For constant A , the method of undetermined coefficients applies when the entries of $\mathbf{F}(t)$ are finite combinations of constants, polynomials, exponentials, sines, and cosines.

Procedure.

1. Solve the associated homogeneous system $\mathbf{X}' = A\mathbf{X}$.
2. Guess the form of $\mathbf{X}_p$ from the form of $\mathbf{F}(t)$.
3. Substitute $\mathbf{X}_p$ into $\mathbf{X}' = A\mathbf{X} + \mathbf{F}(t)$.
4. Solve the resulting algebraic equations for the unknown coefficients.
5. Write $\mathbf{X} = \mathbf{X}_c + \mathbf{X}_p$.

Variation of Parameters for Systems. Let $\Phi(t)$ be a fundamental matrix for the homogeneous system $\mathbf{X}' = A(t)\mathbf{X}$.

$$\mathbf{X}(t) = \Phi(t)\mathbf{C} + \Phi(t) \int \Phi^{-1}(t)\mathbf{F}(t) dt$$

2.6.4 Matrix Exponential

Definition 2.6.8: Matrix Exponential

For an $n \times n$ matrix A , the *matrix exponential* is defined by

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{A^k t^k}{k!}$$

Derivative and Initial Value. The matrix exponential behaves like the scalar exponential:

$$\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A, \quad e^{A \cdot 0} = I$$

Theorem 2.6.9: Homogeneous Solution Using the Matrix Exponential

The solution of $\mathbf{X}' = A\mathbf{X}$, $\mathbf{X}(0) = \mathbf{X}_0$ is

$$\mathbf{X}(t) = e^{At}\mathbf{X}_0$$

More generally, the general solution is

$$\mathbf{X}(t) = e^{At}\mathbf{C}$$

Nonhomogeneous Solution. For constant A , $\mathbf{X}' = A\mathbf{X} + \mathbf{F}(t)$, $\mathbf{X}(t_0) = \mathbf{X}_0$, we have

$$\mathbf{X}(t) = e^{A(t-t_0)}\mathbf{X}_0 + \int_{t_0}^t e^{A(t-s)}\mathbf{F}(s) ds$$

Computing e^{At} by Diagonalization. If $A = PDP^{-1}$, where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ then

$$e^{At} = Pe^{Dt}P^{-1}, \quad e^{Dt} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t})$$

Computing e^{At} by Laplace Transform.

$$\mathcal{L}\{e^{At}\} = (sI - A)^{-1}, \quad e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$$

2.7 Practice Questions

2.7.1 First-Order Differential Equations

Q1 Solve the given differential equation by separation of variables: $e^x y \frac{dy}{dx} = e^{-y} + e^{-7x-y}$

Q2 Find the general solution of the given differential equation: $x^2 y' + x(x+2)y = e^x$

Q3 Solve the given IVP: $(6y + 2t - 3)dt + (8y + 6t - 1)dy = 0$, $y(-1) = 2$

Q4 Solve the given differential equation by finding an appropriate integrating factor: $6xydx + (4y + 9x^2)dy = 0$

Q5 Solve the given differential equation by using an appropriate substitution: $ydx = 2(x+y)dy$

Q6 Solve the given differential equation by using an appropriate substitution: $x \frac{dy}{dx} - (1+x)y = xy^2$

Q7 Solve the given differential equation by using an appropriate substitution: $\frac{dy}{dx} = 4 + \sqrt{y - 4x + 2}$

2.7.2 Higher-Order Differential Equations

Q1 The indicated function $y_1(x)$ is a solution of the following differential equation, find $y_2(x)$:

$$y'' + 2y' + 36y = 0, \quad y_1(x) = xe^{-x}$$

Q2 The indicated function $y_1(x)$ is a solution of the following differential equation, find $y_2(x)$:

$$x^2y'' - 11xy' + 36y = 0, \quad y_1(x) = x^6$$

Q3 Find the general solution of the given equation: $\frac{d^3u}{dt^3} + \frac{d^2u}{dt^2} - 2u = 0$

Q4 Find the general solution of the given equation: $16\frac{d^4y}{dx^4} + 48\frac{d^2y}{dx^2} + 36y = 0$

Q5 Find the general solution of the given equation: $y'' - 2y' + 50y = e^x \cos(7x)$

Q6 Find the general solution of the given equation: $y'' + 3y' + 2y = \frac{1}{2+e^x}$

Q6 Find the general solution of the given equation: $y'' - y = e^{\frac{x}{2}} + 8$

Q7 Find the general solution of the given equation: $y'' + 3y' + 2y = \cos(e^x)$

Q8 Find the general solution of the given equation: $x^2y'' - xy' + y = 14x$

Q9 Find the general solution of the given equation: $16x^2y'' + 16xy' + y = 0$

Q10 Solve the given system of differential equations: $\frac{dx}{dt} = 12x + 19y, \quad \frac{dy}{dt} = x + 6y$

Q11 Find the general solution of the given equation: $y'' + y = \sec(x)$

2.7.3 Series Solutions of Linear Equations

Q1 Find two power series solutions of the given differential equation about the ordinary point $x = 0$:

$$y'' - 4xy' + y = 0$$

Q2 Find the general solution of the given equation: $y'' - xy' - y = 0$

Q3 Find the general solution of the given equation: $x^2y'' + x(x-2)y' + (2-x)y = 0$

2.7.4 The Laplace Transform

Q1 Find the general solution of the given equation: $y'' + 2y' = \delta(t - \pi) - u(t - 2\pi)$

2.7.5 Systems of Linear Differential Equations

Q1 Solve the given initial-value problem:

$$\mathbf{X}' = \begin{pmatrix} \frac{1}{2} & 0 \\ 1 & -\frac{1}{2} \end{pmatrix} \mathbf{X}, \quad \mathbf{X}(0) = \begin{pmatrix} 5 \\ 8 \end{pmatrix}$$

Q2 Find the general solution of the given system:

$$\mathbf{X}' = \begin{pmatrix} 9 & -4 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 9 \end{pmatrix} \mathbf{X}$$

Q3 Find the general solution of the given system: $\frac{dx}{dt} = 5x - y$, $\frac{dy}{dt} = 25x - 5y$

Q4 Find the general solution of the given system: $\frac{dx}{dt} = 8x - y$, $\frac{dy}{dt} = 5x + 4y$

Q5 Find the general solution of the given system:

$$\mathbf{x}' = \begin{pmatrix} 0 & 5 \\ -1 & 6 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 16 \\ -16 \end{pmatrix} e^t$$